Optimisation

1

a

i $(-1, 96)$ ii 96 iii -120

b

i No stationary points in the given domain.

ii 88

iii -12

c

i $(-2, -128)$ ii 72 iii -128

d

i No stationary points in the given domain.

ii 28

iii -80

e

i $\left(\frac{26}{3}, -\frac{319}{27}\right), (4, 39)$ ii 339

iii $-\frac{319}{27}$

2

a $(1, -2)$ b -2

3

a Absolute minimum, because the parabola is concave up as seen by the positive coefficient of x^2.

b $y = 2$

4

a Absolute maximum, because the parabola is concave down as seen by the negative coefficient of x^2.

b $y = \frac{1}{2}$

Applications

5

a $t = 3$

b 122 units

6

a $\frac{dy}{dx} = \frac{x}{50} - 1$

b $x = 50$

c 25 units

7

a $\frac{dy}{dx} = \frac{7}{20} - \frac{x}{2}$

b $x = 0.7$

c $0.12\ \mathrm{kg/m^2}$

8

a $y = x\,(24 - x)$

b $x = 12$

c 144

9

a $A = x\,(20 - x)$

b 10 units

c 10 units

10

a $(168 - x)\ \mathrm{m}$

b $x\,(168 - x)\ \mathrm{m}^2$

c $84\ \mathrm{m}$

d $7056\ \mathrm{m}^2$

11

a $t = 2.84$ s

b $t = 1.38$ s

c $34.25\ \mathrm{m}$

12

a $V' = -544\,500 + 99\,000t - 4500t^2$

b $-4500\ \mathrm{L/min}$

c $t = 0\ \mathrm{min}$

13

a 128 students

b 451 students

c $t = 0, t = \sqrt{31}$

d 1089 students

e $t = \sqrt{\frac{31}{3}}$

14

a $L = 28 - 2W$

b $A = 28W - 2W^2$

c $7\ \mathrm{m}$

d $14\ \mathrm{m}$

15

a $h = 48 - 4w$

b $V = 144w^2 - 12w^3$

c $w = 8 \text{ cm}$

d 3072 cm^2

e $24 \text{ cm} \times 8 \text{ cm} \times 16 \text{ cm}$

16

a $72 - 2w$

b $45 - 2w$

c $4w^3 - 234w^2 + 3240w$

d $w = 9 \text{ cm}, w = 30 \text{ cm}$

e $w = 9 \text{ cm}$

f $54 \text{ cm} \times 27 \text{ cm} \times 9 \text{ cm}$

17

a $V = 4x^3 - 264x^2 + 3780x$

b $x = 9 \text{ cm}$

c $15\,552 \text{ cm}^3$

18

a $h = \dfrac{256}{x^2}$

b $S = x^2 + \dfrac{1024}{x}$

c $x = 8$

19

a $h = \dfrac{5 - 2r^2}{2r}$

b $V = \dfrac{5\pi r - 2\pi r^3}{2}$

c $r = \sqrt{\dfrac{5}{6}}$

20

a $h = \dfrac{600}{\pi r^2}$

b $A = \dfrac{1200}{r} + 2\pi r^2$

c $r = \sqrt[3]{\dfrac{300}{\pi}}$

d 394 cm^2

e 9 cm

21

a $l = \dfrac{8}{w}$

b $\dfrac{dl}{dw} = -8w^{-2}$

c No

d 10 m

22

a $r = 8 - \dfrac{8}{13}h$

b $h = \dfrac{13}{3} \text{ cm}$

23

a $y = 3 - \dfrac{3x}{4}$

b $A = 3x - \dfrac{3x^2}{4}$

c $\dfrac{dA}{dx} = 3 - \dfrac{3x}{2}$

d 1.53 m^2

24

a $y = 8 - \frac{\pi x}{4}$

b $A = 8x - \frac{\pi x^2}{8}$

c $\frac{dA}{dx} = 8 - \frac{\pi x}{4}$

d $x = 10.2$

e $x = 8.9$

f 40.1 m^2

25

a $V = \frac{x}{3}\pi\left(2mx - x^2\right)$

b $x = \frac{4m}{3}$

26

a $11 - 0.25x$

b $1800 + 180x$

c $R = 19\,800 + 1530x - 45x^2$

d $x = 17$

e $\$6.75$

f 4860

27

a No solution provided.

b $w = 3$

c $d = 3\sqrt{2}$

d $S = 54k$

e $S = \frac{81\sqrt{6}}{4}k$

f 8.9%